1st Order DEs: Integrating Factors

Questions 1–45

Question 1. Find

a)
$$\frac{\mathrm{d}}{\mathrm{d}x}(6xy)$$

b)
$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{3x}y)$$

c)
$$\frac{\mathrm{d}}{\mathrm{d}x}(y\tan 4x)$$

d)
$$\frac{\mathrm{d}}{\mathrm{d}x}(y\ln 2x)$$

Question 2.

a) Find
$$\frac{\mathrm{d}}{\mathrm{d}x}(yx^2)$$

b) Find the general solution of each of the following differential equations

(i)
$$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=4$$

(ii)
$$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=4x$$

Question 3.

a) Find

$$\frac{\mathrm{d}}{\mathrm{d}x}(y \sin x)$$

b) Find the general solution of each of the following differential equations

(i)

$$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} + y \cos x = 3x^2$$

(ii)

$$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} + y \cos x = e^x$$

Question 4.

a) Find

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{5x} y)$$

b) Find the general solution of the differential equation

$$e^{5x} \frac{\mathrm{d}y}{\mathrm{d}x} + 5e^{5x} y = 6e^{3x}$$

c) Hence, given that $y = 0$ when $x = \ln 2$, show that the particular solution of this differential equation can be written as

$$y = \frac{2e^{3x} - 16}{e^{5x}}$$

Question 5. The following differential equations are such that the left-hand side can be written in the form

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y)$$

without the need to rearrange and find an integrating factor. Find the general solution to each of the differential equations

a)

$$x^4\frac{\mathrm{d}y}{\mathrm{d}x} + 4x^3y = x^8$$

b)

$$\cos 2x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\sin 2x = \cos 2x$$

c)

$$\tan 3x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y\sec^2 3x = \cos 3x$$

d)

$$e^{-9x}\frac{\mathrm{d}y}{\mathrm{d}x} - 9e^{-9x}y = e^{-8x}$$

Question 6. Find and simplify

a)

$$e^{\int 2x\mathrm{d}x}$$

b)

$$e^{\int \frac{1}{x}\mathrm{d}x}$$

c)

$$e^{\int \frac{1}{2x+1}\mathrm{d}x}$$

d)

$$e^{\int \tan x\mathrm{d}x}$$

Question 7. For each of the following differential equations, find the integrating factor. You are not required to solve the differential equation.

a)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + 8y = e^x$$

b)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = x$$

c)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - xy = 4$$

d)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + (2x+1)y = x^2$$

Question 8. It is given that
$$x^3\frac{\mathrm{d}y}{\mathrm{d}x} + x^3y = \frac{2}{e^x}$$

a) Show that an integrating factor is e^x
b) Solve the differential equation using the integrating factor in a).
c) Solve the differential equation using the following integrating factors. In each case, explain why your answer is equivalent to the answer given in b).
   (i) $2e^x$
   (ii) e^{x+2}

Question 9. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$$

where $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ are continuous functions of x. Suppose that the equation is multiplied by a function $\mu(x)$ so that the left-hand side becomes the derivative of a product:

$$\mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu(x)\mathrm{P}(x)y = \frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y)$$

a) Use the product rule to show that $\mu(x)$ must satisfy

$$\mu'(x) = \mathrm{P}(x)\mu(x)$$

b) Hence show that a suitable integrating factor is

$$\mu(x) = e^{\int \mathrm{P}(x)\mathrm{d}x}$$

Question 10. Archie is attempting to solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = e^{3x}$$

His attempt is shown below: Integrating Factor $= e^{\int -3\mathrm{d}x} = -3e^{-3x}$
Multiplying the equation by $-3e^{-3x}$:

$$-3e^{-3x}\frac{\mathrm{d}y}{\mathrm{d}x} + 9e^{-3x}y = -3$$

This gives: $\frac{\mathrm{d}}{\mathrm{d}x}(-3e^{-3x}y) = -3$
Integrating: $-3e^{-3x}y = -3x + c$
Rearranging: $y = e^{3x}\left(x - \frac{c}{3}\right)$

a) Identify the error in Archie's work.
b) Verify whether Archie's integrating factor $-3e^{-3x}$ is valid
c) A different student uses the integrating factor e^{-3x} and obtains the solution $y = e^{3x}(x + c)$. Explain why this solution is equivalent to the solution found in Archie's work.

Question 11. Find the general solution for each of the following differential equations:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 10y = e^{2x}$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 7y = 8$$

c)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 8y = 100$$

d)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 20y = 6e^{-12x}$$

Question 12. Simplify the following:

a)

$$e^{\ln x}$$

b)

$$e^{\ln x^2}$$

c)

$$e^{4\ln x}$$

d)

$$e^{-3\ln x}$$

Question 13. Find the general solution for each of the following differential equations:

a)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = x$$

b)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 30$$

c)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3y}{x} = x^3$$

d)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5y}{x} = \frac{40}{x^4}$$

Question 14. Find the general solution for each of the following differential equations:

a)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} = 6x$$

b)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x-3} = 12$$

c)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4y}{2x+1} = 2$$

Question 15. It is given that

$$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = r$$

where p, q and r are positive constants. It is also given that $y = \frac{2r}{q}$ when $x = 0$

a) Find y in terms of x

b) Find the value of y as $x \to \infty$

Question 16. Find the general solution to

a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1$$

b)

$$(x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + y = 2$$

c)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 1$$

d)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 2x^3$$

Question 17. Find the general solution to

a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = \sqrt{x}$$

b)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \frac{1}{\sqrt{x}}$$

Question 18. Find the general solution to

a)

$$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + x^{\frac{3}{2}}y = 2$$

b)

$$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{\frac{3}{2}}y = 2$$

Question 19. It is given that $x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 7xy = \frac{2}{x^5}$ where $x > 0$. Given that $y = 1$ when $x = 2$, show that y is a decreasing function.

Question 20.

a) Find the general solution of the differential equation $\sin x\frac{\mathrm{d}y}{\mathrm{d}x} - y\cos x = 2^x \sin^2 x$ where $x \in \mathbb{R}$

b) Hence find the particular solution of this differential equation, given that $y = 0$ when $x = \frac{\pi}{2}$

Question 21. The following questions in this section require the use of A-Level Maths Calculus techniques, including Reversing the Chain Rule, Double Angle Formulae, Integration by Parts
Find the general solution for each of the following differential equations:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + xy = x$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 6x^3y = 8x^3$$

Question 22. Find the general solution for each of the following differential equations:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x = \cos x$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin 2x = 4\sin 2x$$

Question 23. Find the general solution to

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 6\sqrt{x^3 + 2}$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = \frac{6}{\sqrt{x^4 + 2}}$$

Question 24. It is given that the curve that satisfies the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{6y}{3x - 2} = 54(3x - 2)^6$$

has a stationary point when $x = \frac{1}{3}$
Find the equation of the curve.

Question 25. Find y in terms of x, given that

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 36x^6 \ln x$$

and $y = 2$ when $x = 1$

b)

$$x\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sqrt{x} = 8x\sqrt{x^3 + 4x}$$

and $y = 1$ when $x = \sqrt{5}$

c)

$$\tan x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec^2 x = \tan^2 x$$

and $y = 1$ when $x = \frac{\pi}{4}$

Question 26. The gradient at any point (x, y) on a curve C is proportional to $(y - x)$. The curve passes through $(1, 3)$ and the gradient of the curve at this point is -8. Find the equation of the curve.

Question 27.

a) Find the general solution of the differential equation

$$(1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x=2(\sin x+\cos x)$$

b) Hence, given that $y=5$ when $x=\frac{\pi}{2}$, show that the particular solution of this differential equation can be written as

$$y=\frac{\sin x-\cos x+4}{\sin^2 x}$$

Question 28.

a) Find the general solution of the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x}+(2x+1)y=xe^x$$

b) Hence, given that $y=\frac{1}{3}$ when $x=1$, show that the particular solution of this differential equation can be written as

$$y=\frac{(3x-1)e^x+e^{3-2x}}{9x}$$

Question 29. Given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = \sin 2x$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and $y = \frac{4}{3}$ when $x = 0$, find y in terms of x

Question 30.

a) Find the general solution of the differential equation

$$\sec x\frac{\mathrm{d}y}{\mathrm{d}x} + y\operatorname{cosec} x = 2\tan x$$

where $0 < x < \frac{\pi}{2}$

b) Hence find the particular solution of this differential equation, given that $y = \frac{12}{\sqrt{3}}$ when $x = \frac{\pi}{3}$

Question 31. Given that

$$x^4\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{x^3y}{\ln x}=3$$

where $x>1$, and $y=e^{-3}$ when $x=e$, show that

$$y=\frac{7e^{-3}x^3-3\ln x-1}{3x^3\ln x}$$

Question 32. The following questions in this section require the use of A-Level Further Maths Calculus techniques, including Partial Fractions, Inverse Trigonometric Functions, Hyperbolic Functions, Inverse Hyperbolic Functions

Find the general solution for each of the following differential equations, giving your answer in the form $y=\mathrm{f}(x)$:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x}+y\sinh x=2\sinh x$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x}-2y\cosh 3x=6\cosh 3x$$

c)

$$2\cosh^2 x\frac{\mathrm{d}y}{\mathrm{d}x}+y\sinh 2x=\sinh^5 x\cosh^2 x$$

Question 33. Find the general solution of the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = \frac{7-x}{(x^2+7)(x+1)}$$

where $x \neq 0$ and $x \neq -1$

Question 34.

a) Show that

$$e^{\int \frac{1}{1-x^2}\,\mathrm{d}x} = \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$

It is given that

$$(1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x} + y = (1-x)^{\frac{3}{2}}(1+x)$$

where $-1 < x < 1$

b) Find the general solution of the differential equation, given that $y = \frac{2}{3}$ when $x = 0$

Question 35. Given that

$$(9-4x^2)\frac{\mathrm{d}y}{\mathrm{d}x}-4xy=2$$

where $-\frac{3}{2}<x<\frac{3}{2}$, and $y=0$ when $x=\frac{3\sqrt{2}}{4}$, find y in terms of x

Question 36. Given that

$$2x\frac{\mathrm{d}y}{\mathrm{d}x}-y=\sqrt{\frac{4x^3}{4x-x^2-1}}$$

where $2-\sqrt{3}<x<2+\sqrt{3}$, and $y=\sqrt{\frac{7}{2}}$ when $x=\frac{7}{2}$, find y in terms of x

Question 37. Prerequisite knowledge for Q37-Q38: Complementary Functions for solving Non-Homogeneous Differential Equations

Solve the following differential equations by finding the complementary solution and a particular integral.

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = e^{5x}$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = x$$

c)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 17\sin x$$

d)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = e^{7x}$$

Question 38.

a) Show that $y = Ax + B + C\sin x + D\cos x$ is a particular integral of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 15\cos x - 4x$$

b) Hence find the general solution of the differential equation.

Substitutions

Question 39. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = xy^2$$

where $x > 0$. Use the substitution $u = \frac{1}{y}$ to find the general solution for y in terms of x

Question 40. It is given that

$$5xy^4\frac{\mathrm{d}y}{\mathrm{d}x} + 3y^5 = \frac{\sin 2x}{x^2}$$

where $x > 0$. Use the substitution $u = y^5$ to find the general solution for y in terms of x

Question 41. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4xy^3 = y$$

Given that $y = \frac{1}{3}$ when $x = 0$, use the substitution $u = \frac{1}{y^2}$ to find a particular solution for y^2 in terms of x

Question 42. It is given that

$$x^3 e^y \frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 e^x - 3x^2 e^y$$

Given that $y = \ln 6$ when $x = 1$, use the substitution $u = e^y$ to find a particular solution for y in terms of x

Question 43. It is given that

$$x\frac{\mathrm{d}y}{\mathrm{d}x} = y(\ln y + x)$$

where $x > 0$ and $y > 0$. Use the substitution $u = \ln y$ to find the general solution for y in terms of x

Question 44. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = e^x \cos^2 y + \sin y \cos y$$

where $x \in \mathbb{R}$. Use the substitution $u = \tan y$ to find the general solution for y in terms of x

Question 45. For $x > 0$ and $y > 0$, it is given that

$$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = ry^{1-n}$$

where p, q, r and n are positive constants.

a) Use the substitution $u = y^n$ to find the general solution for y in terms of x

b) Find the value of y as $x \to \infty$ in terms of the constants q, r and n